Scrubgnome Away from Home

kultrshok

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Twig's pocket edition. Strip the snitchy metadata out of phone videos before you post them.

> "Great gears! Your video is blabbing where, when and with what you filmed it. Hand it over."
> — Twig

Scrubgnome Away from Home (AFH) is the Android sidekick to desktop Scrubgnome. It does one job fast:

1. **Pick** a video (or *Share* one to "Scrub with Twig" from your gallery).
2. **See** what it's leaking: GPS location, recording time, phone make/model, software version, hidden data tracks.
3. **Scrub & save** a clean copy to `Movies/Scrubgnome/` with a random filename.
4. **Share / upload** the clean copy.

Optional: **Listen for hidden tracking tones.** Twig scans the audio for inaudible "ultrasonic beacon" signals (17.5–22 kHz) and warns you if he hears any.

Privacy promise: the app has **no internet permission at all**. Android won't let it talk to the network even if it wanted to. Everything happens on your phone.

Install (no app store needed)

1. Open the repo's **Actions** tab → newest green **Build APK** run → download **ScrubgnomeAFH-apk** (a zip holding the APK). Tagged versions also show up under **Releases**.
2. Copy the APK to your phone and tap it. Android will ask you to allow installs from that app (Files/Chrome). Allow it once.
3. Requires **Android 10 or newer**.

What gets removed

Leak	Where it hides	What Twig does
GPS location	`©xyz`, QuickTime location keys, XMP	box neutralized and zeroed
Recording date/time	`mvhd` / `tkhd` / `mdhd` headers, `©day`, XMP	timestamps zeroed
Phone make & model	`com.android.model`, `©mak`/`©mod`, Apple keys	box neutralized and zeroed

Leak	Where it hides	What Twig does
OS / software version	`com.android.version`, `©swr`, `©too`	box neutralized and zeroed
Hidden data tracks	Pixel `mett`, iPhone `mebx`, camera-motion/GPS, subtitle tracks	sample bytes zeroed **and** track removed
XMP sidecar blobs	top-level `uuid` boxes	neutralized and zeroed
Vendor trailers	junk after the movie data (e.g. Samsung)	cut off
Handler names	`hdlr` (“VideoHandle”, “Core Media…”)	blanked
Original filename	`PXL_20261108_001736.mp4` *is* a timestamp	clean copy gets `scrubbed_<random>.mp4`

Lossless: nothing is re-encoded. Metadata boxes become same-size `free` boxes full of zeros, so every pointer in the file stays valid and the video and sound are bit-for-bit untouched. After scrubbing, Twig re-scans the result and refuses to save if anything survived.

Supported: MP4, MOV, 3GP (everything phones record). Not supported: WebM/MKV.

The hidden-tone check (and its limits)

Cross-device ad tracking sometimes plays near-ultrasonic beacons (steady pilot tones or short data bursts) that phones can hear and people can’t. Twig decodes the audio on-device, runs an FFT over 17.5–22 kHz and reports:

Result	Meaning
🔴 **Steady inaudible tone**	A constant narrow tone, beacon-like. Could also be electronics (chargers, displays, pest repellers).
🟠 **Possible data bursts**	Narrow blips that come and go or hop frequencies.
🔵 **Some ultrasonic noise**	Broadband hiss, no beacon pattern.
🟢 **No tracking tones** / **No room for hidden tones**	Clean, or the phone’s encoder already cut everything above the beacon band.

It’s a **warning, not proof**. It can miss a very faint beacon buried under loud full-range sound, and it can’t detect watermarks hidden *inside* the audible range. The pocket edition doesn’t remove tones; use desktop Scrubgnome’s audio cleaning for that.

Build it yourself

Easiest: push to GitHub and let Actions build it (see `.github/workflows/build-apk.yml`).

Android Studio: *File → Open* this folder, wait for Gradle sync, press ▶ with your phone plugged in (USB debugging on).

Desktop self-test of the scrubber core (needs Java 17+ and ffmpeg):

```
bash tools/selftest.sh
```

Project layout

```
app/src/main/java/com/kultrshok/scrubgnome/
├── core/Mp4Scrubber.java         ← the scrubber (pure Java, no Android; reusable module)
├── core/UltrasonicAnalyzer.java  ← the tone detector (pure Java, no Android; reusable module)
├── AudioToneScanner.java         ← Android audio decoder that feeds the analyzer
└── MainActivity.java             ← the one-screen UI (Twig's voice lives here)
tools/                            ← desktop self-test
```

No third-party libraries. The two `core/` classes drop straight into any Java/Kotlin project.

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